

COURSE TITLE	: MEDICAL ELECTRONICS LAB
COURSE CODE	: 5248
COURSE CATEGORY	: A
PERIODS/WEEK	: 4
PERIODS/SEMESTER	: 52
CREDITS	: 3

LIST OF EXPERIMENTS

On completion of the course the student will be able:

- 1.1 To setup a Bio amplifier circuit using op-amp and find out CMRR.
- 1.2 To setup a Phase detector circuit using PLL IC and observe the phase difference between input and output.
- 1.3 To setup a Notch filter circuit using op-amp and
 - (i) To plot frequency response
 - (ii) To measure the band reject frequency.
- 1.4 To demonstrate IC 4051 and verify the output.
- 1.5 To setup a Threshold detector using op-amp for a given voltage level.
- 1.6 To plot Thermistor characteristics curve.
- 1.7 To measure skin contact impedance.
- 1.8 To plot LDR characteristic curve.
- 1.9 To setup a Band pass filter and plot the frequency response.
- 1.10 To setup a Flash ADC (3 bit) circuit.
- 1.11 To setup a Precision rectifier circuits using op-amp (Half wave and Full wave).
- 1.12 To setup a patient isolation circuit using opto-coupler (IC MCT2E).
- 1.13 To setup a Sample and hold circuit using op-amp.
- 1.14 To demonstrate AD 590 and plot the characteristics.
- 1.15 To demonstrate the characteristics of different electrodes such as ECG, EEG and EMG.